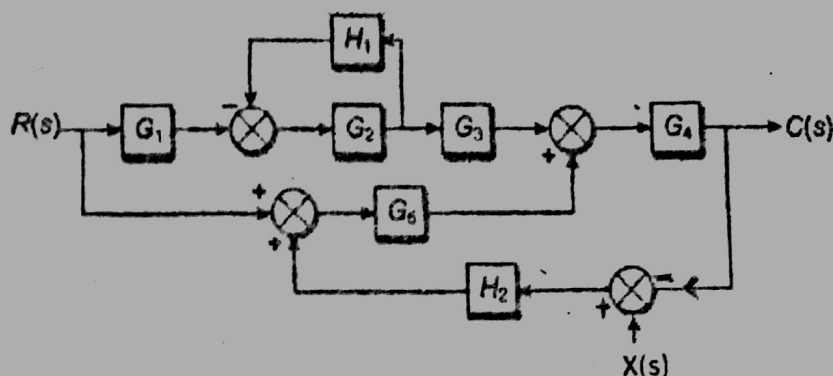


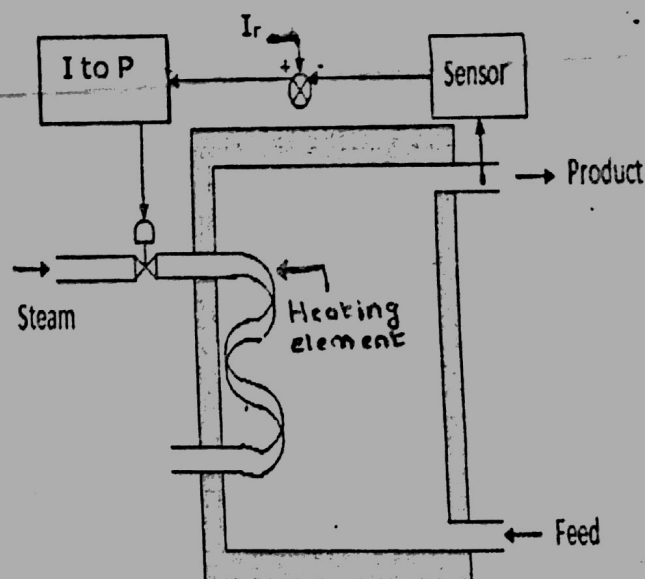
- Q1 The block diagram of a system is shown below. Using the block diagram reduction techniques, find all the possible transfer functions. [10]



- Q2 A temperature control industrial heating system is shown in figure. Modeling information is as below:

- Steam flow rate = Q m³/sec
- heat produced by the heating element is directly proportional to steam flow rate. (Take constant of proportionality as $K_h = 1$.)
- Feed Temperature (T_i), Product temperature (T)
- Thermal capacitance of the tank (C) = 5 J/°C
- Thermal resistance of tank (R_t) = 2 °C/W
- Temperature sensor is a first order system with transfer function $\frac{K_t}{\tau_t s + 1}$

where $K_t = 5 \text{ mA}/^\circ\text{C}$, $\tau_t = 2 \text{ sec}$.



- I to P is linear current-to-pressure converter and converts 0-16 mA to 0-12 psi pressure.
- Control valve is linear with constant $K_v = 1 \text{ m}^3\text{sec}^{-1}\text{psi}^{-1}$

i. Draw the block diagram of entire system showing transfer function of each block.

ii. Derive the overall transfer function $\frac{\Delta T(s)}{\Delta I_r(s)}$.

iii. Determine the sensitivity of the closed loop transfer function with respect to K_t for $\omega = 10 \text{ rad/s}$.

iv. Determine the steady-state error for unit step input.

[15]

- Q.3 The open loop transfer function of negative feedback is given below. Draw the Nyquist plot and find the range or value of K for closed loop system stability. [18]

$$G(s)H(s) = \frac{K(s+1)}{s(s-1)(s+10)}$$

- Q4. The plant transfer function of a system is $\frac{4}{2s+1}$. A proportional (gain K) and integral (gain K_i) controller is added in the system so that the damping ratio is 0.5 and settling time (corresponding to 2% tolerance band) is less than 0.1 second. For this system,
- Find range of K and K_i
 - with values obtained in part (a), find range of steady state error for unit ramp input. [15]

- Q5. The characteristic equation of a negative feedback system is given below. Sketch the root locus of this system clearly showing all the steps involved. $0 < K < \infty$. [15]

$$(s^2+2s)(s^2+2s+5) + Ks + K = 0$$

- Q.6 The open loop transfer function of negative feedback is given below. Sketch the bode plots and therefrom determine the range of K for closed loop system to be stable. **Start your plots at $K=1$ and $\omega = 1$ rad/s.** [17]

$$G(s)H(s) = \frac{100K(s+5)(s+40)}{s^3(s+100)(s+200)}$$

$s^2 + 300s +$